Waiting-Line Models

Module 8 — Queuing Theory & Capacity Economics



MGT 207 — Production and Operations Management

Common Queuing Situations

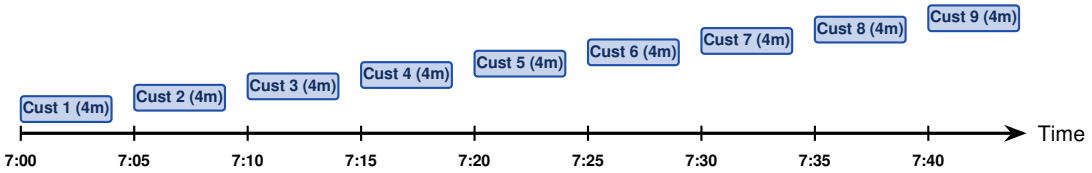
TABLE D.1 Common Queuing Situations

SITUATION	ARRIVALS	QUEUE	SERVICE FACILITY
Supermarket	Grocery shoppers	Checkout lines	Cashiers
Highway toll booth	Autos	Highway lane	Toll taker / machine
Doctor's office	Patients	Waiting room	Doctor
Computer system	Programs to be run	Data storage	CPU
Telephone company	Callers	Switching equipment	Trunk lines / operators
Bank	Customers	Waiting line	Teller
Machine maintenance	Broken machines	Machines waiting for repair	Repairperson
Harbor	Ships and barges	Waiting in port	Docks / unloading crew

The Impact of Variability

Deterministic Benchmark (Zero Variability)

Question: Customers arrive once every **5 minutes** and service takes **4 minutes**. How long does a customer wait? $\Longrightarrow$ **0 minutes!**

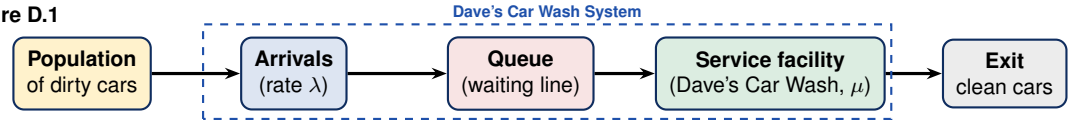


Arrival Bunching & Variability

- **Question:** Given 12 customer arrivals per hour, think about a situation where **12 customers arrive altogether at the top of each hour**. What's the **longest time** a customer has to wait in line?
- **Answer:** Wait times are $0, 4, 8, \ldots, 44$ min $\Longrightarrow$ **Longest wait** $=$ **44** minutes (12th customer; average wait $=$ **22** min).

Parts of a Waiting Line

Figure D.1



1. Arrival Characteristics

- **Population size:** Unlimited (infinite) vs. Limited (finite)
- **Behavior:** Wait, *balk* (refuse to join), or *renege* (leave line)
- **Pattern:** Scheduled vs. Random (**Poisson** distribution)

2. Waiting-Line Features

- **Queue length:** Limited vs. Unlimited line capacity
- **Queue discipline:** First-In, First-Out (**FIFO**) vs. Priority rules (e.g., ER triage)

3. Service Characteristics

- **Design:** Single vs. Multiple channels (servers); Single vs. Multiple phases
- **Distribution:** Constant vs. Random (**Negative exponential**)

Poisson Distribution

Formula for Random Arrivals

Used to describe the number of independent arrivals per unit of time:

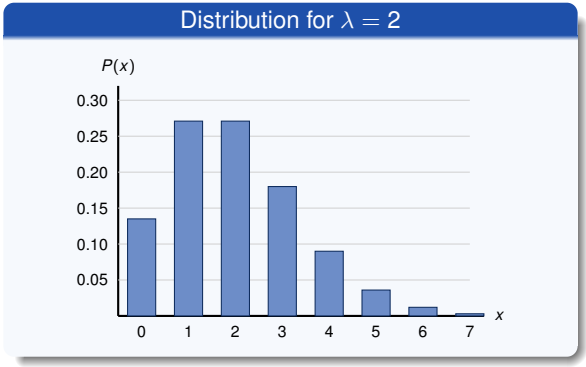
$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \quad \text{for } x = 0, 1, 2, 3, 4, \ldots$$

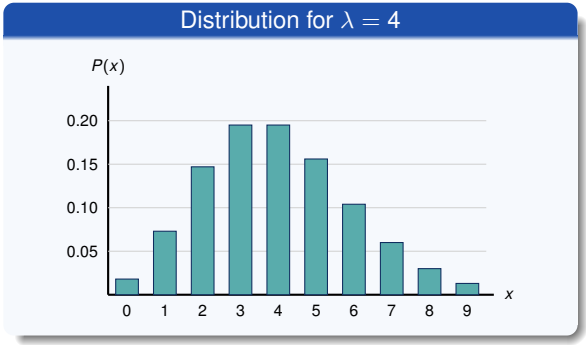
Parameter Definitions

- $P(x) =$ probability of x arrivals
- $x =$ number of arrivals per unit of time
- $\lambda =$ average arrival rate
- $e \approx 2.7183$ (base of natural logarithms)
- $x! = x(x-1)(x-2)\cdots(2)(1)$

Poisson Distribution

Figure D.2 Two Examples of the Poisson Distribution for Arrival Times





Negative Exponential Distribution

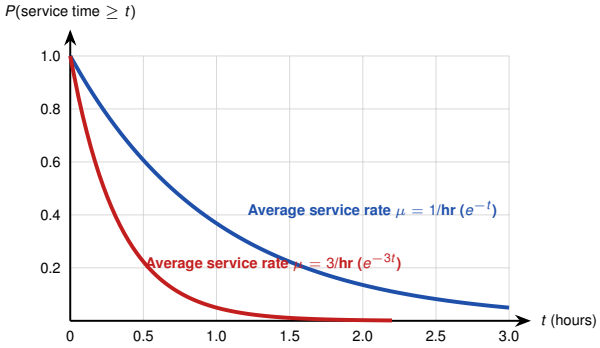
Continuous Service Time Model

Describes the probability that service time (or interarrival time) is greater than or equal to t:

$$P(\text{service time} \geq t) = e^{-\mu t} \quad \text{for } t \geq 0$$

- μ = average number of customers served per period
- t = length of service time
- $e \approx 2.7183$
- **Connection:** If arrivals follow a Poisson process with rate λ, interarrival times follow an Exponential distribution with mean $1/\lambda$.

Figure D.4 Negative Exponential Distribution



Measuring Queue Performance

Queuing models help managers evaluate seven key operating characteristics:

Time & Line Length Metrics

1. **Average time in the queue (W_q):** Average time each customer spends waiting in line.
2. **Average queue length (L_q):** Average number of customers waiting in the line.
3. **Average time in the system (W_s):** Waiting time plus service time ($W_q + 1/\mu$).
4. **Average number in the system (L_s):** Customers waiting plus those being served.

Facility & Probability Metrics

5. **Probability facility is idle (P_0):** Probability that 0 units are in the system.
6. **Utilization factor (ρ):** Percentage of time the service facility is busy.
7. **Probability of n units in system (P_n):** Likelihood of a specific congestion level (e.g., $P_{n>k}$).

Queuing Models

Shared Characteristics Across All Four Models

The four queuing models described in this chapter all share three core characteristics:

1. Poisson Arrivals

Random, memoryless arrivals described by the Poisson distribution (λ).

2. FIFO Discipline

First-In, First-Out queue rule with no balking or reneging.

3. Single Service Phase

Customers complete service in a single stage and exit the system.

Queuing Models

TABLE D.2 Queuing Models Described in This Chapter

MODEL / NAME	EXAMPLE	CHANNELS	PHASES	ARRIVALS	SERVICE TIME	POPULATION	DISCIPLINE
A: Single-server system (M/M/1)	Information counter at department store	Single	Single	Poisson	Negative exponential	Unlimited	FIFO

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B: Multiple-server system (M/M/S)	Airline ticket counter	Multiple	Single	Poisson	Negative exponential	Unlimited	FIFO
C: Constant-service (M/D/1)	Automated car wash	Single	Single	Poisson	Constant	Unlimited	FIFO

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B: Multiple-server system (M/M/S)	Airline ticket counter	Multiple	Single	Poisson	Negative exponential	Unlimited	FIFO
C: Constant-service (M/D/1)	Automated car wash	Single	Single	Poisson	Constant	Unlimited	FIFO
D: Finite-population	Shop with only a dozen machines that might break	Single	Single	Poisson	Negative exponential	Limited	FIFO

Model A — Single-Server (M/M/1)

Arrival & Queue Discipline Assumptions (1 of 2)

1. **FIFO Discipline (No Balking/Reneging):** First-in, first-out service; every arrival waits regardless of line length.
2. **Stationary, Independent Arrivals:** Arrival events are independent with a constant mean rate λ.
3. **Poisson from Infinite Population:** Arrivals follow a Poisson process drawn from a very large population.

Service Time & Stability Assumptions (2 of 2)

4. **Independent Service Times:** Customer service durations are mutually independent with known mean rate μ.
5. **Exponential Service Distribution:** Service times follow the negative exponential distribution.
6. **Stability Condition ($\mu > \lambda$):** Average service rate exceeds arrival rate ($\rho = \lambda/\mu < 1$).

A Clinic / Repair Shop

M/M/1 Simulation & Analytical Example

A Repair Shop

Problem Setup

- On average, cars arrive at a repair shop at the rate of $\lambda = \mathbf{2}$ cars per hour (mean interarrival time $= 30$ min).
- A single mechanic is able to serve on average $\mu = \mathbf{3}$ cars per hour (mean service time $= 20$ min).

Simulation Exercise before Closed-Form Equations

Construct an Excel Monte Carlo simulation model to estimate how long an average car has to wait in line (W_q) to see the mechanic.

A Repair Shop: Excel Spreadsheet Simulation Setup

Generating Exponential Interarrival & Service Times in Excel

Column	Variable	Excel Formula / Logic
Col B	Car interarrival time (min)	$= -(60/\lambda) * \texttt{LN(RAND())}$
Col C	Car arrival clock time	= Prev. car arrival time + Interarrival time
Col D	Time service begins	= Car arrival time + MAX(Prev. exit time − Arrival time, 0)
Col E	Length of service visit (min)	$= -(60/\mu) * \texttt{LN(RAND())}$
Col F	Car exit clock time	= Time service begins (Col D) + Service length (Col E)
Col G	Car wait time in queue (W_q)	= Time service begins (Col D) − Car arrival time (Col C)
Col I	Average car wait time	= AVERAGE(Wait times across n simulated cars)

Model A — Single-Server Queuing Formulas (1 of 3)

TABLE D.3 Queuing Formulas for Model A: Single-Server System (M/M/1)

Parameters & System Performance (L_s and W_s)

- $\lambda =$ mean number of arrivals per time period
- $\mu =$ mean number of people or items served per time period

1. **Average number of units (customers) in the system (L_s)** *(waiting and being served)*:

$$L_s = \frac{\lambda}{\mu - \lambda}$$

2. **Average time a unit spends in the system (W_s)** *(waiting time plus service time)*:

$$W_s = \frac{1}{\mu - \lambda}$$

Model A — Single-Server Queuing Formulas (2 of 3)

TABLE D.3 Queuing Formulas for Model A: Single-Server System (M/M/1)

Queue Length (L_q), Queue Wait (W_q), and Utilization (ρ)

3. **Average number of units waiting in the queue (L_q)**:

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

4. **Average time a unit spends waiting in the queue (W_q)**:

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{L_q}{\lambda}$$

5. **Utilization factor for the system (ρ)** *(probability the service facility is being used)*:

$$\rho = \frac{\lambda}{\mu}$$

Model A — Single-Server Queuing Formulas (3 of 3)

TABLE D.3 Queuing Formulas for Model A: Single-Server System (M/M/1)

Idle Probability (P_0) and Tail Congestion Probability ($P_{n>k}$)

6. **Percent idle time (P_0)** *(probability no one is in the system)*:

$$P_0 = 1 - \frac{\lambda}{\mu}$$

7. **Probability that the number of units in the system exceeds k ($P_{n>k}$)**:

$$P_{n>k} = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

Single-Server Example ($\lambda = 2$ cars/hr, $\mu = 3$ cars/hr)

Queue Wait Time (W_q)

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{2}{3(3-2)} = \mathbf{\frac{2}{3}} \text{ hour}$$

$$\implies \textbf{40} \text{ minutes average wait in line}$$

Utilization (ρ) & Idle Probability (P_0)

$$\rho = \frac{\lambda}{\mu} = \frac{2}{3} = \textbf{0.667} \ (\textbf{66.7}\% \text{ busy})$$

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{2}{3} = \textbf{0.333} \ (\textbf{33.3}\% \text{ idle})$$

Managerial Insight

Even though the mechanic is idle **33.3%** of the time, random arrivals and service times cause the average car to wait **40 minutes** before service even begins!

Single-Server Example ($\lambda = 2$ cars/hr, $\mu = 3$ cars/hr)

Cars in System (L_s)

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{2}{3-2}$$

$$= \mathbf{2} \text{ cars}$$

(in system on average)

Time in System (W_s)

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{3-2}$$

$$= \mathbf{1} \text{ hour } (\mathbf{60} \text{ min})$$

(40m wait + 20m service)

Cars in Queue (L_q)

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{2^2}{3(3-2)}$$

$$= \frac{4}{3} = \mathbf{1.33} \text{ cars}$$

(waiting in line on average)

Single-Server Example: Congestion Probabilities

Probability of More Than k Cars in the System $P_{n>k} = \left(\frac{2}{3}\right)^{k+1}$

k	$P_{n>k} = (2/3)^{k+1}$	INTERPRETATION
0	**0.667**	← Note that this equals $1 - P_0 = 1 - 0.333$ (server busy)
1	0.444	44.4% chance at least 1 car is waiting in the queue
2	0.296	29.6% chance more than 2 cars are in the system
3	**0.198**	← **19.8% chance** that more than 3 cars are in the system
4	0.132	13.2% chance more than 4 cars are in the system
5	0.088	8.8% chance more than 5 cars are in the system
6	0.058	5.8% chance more than 6 cars are in the system
7	0.039	3.9% chance more than 7 cars are in the system

Single-Channel Economics (Repair Shop Cost Analysis)

Economic Parameters (8-Hour Day)

- Arrival rate: $\lambda = 2$ cars/hr $\implies$ **16** cars/day
- Average queue wait: $W_q = \frac{2}{3}$ hr (40 min)
- Customer waiting cost: \$**15** per hour of wait
- Mechanic daily wage: \$**88** per day (1 mechanic)

Total Daily Cost Calculation

- **Total daily customer wait time:**

$$16 \text{ cars/day} \times \frac{2}{3} \text{ hr/car} = \mathbf{10\frac{2}{3}} \text{ hours/day}$$

- **Total daily waiting cost:**

$$10\frac{2}{3} \text{ hrs} \times \$15/\text{hr} = \$\mathbf{160}/\text{day}$$

- **Total daily system cost:**

$$\$160 \text{ (waiting)} + \$88 \text{ (service)} = \$\mathbf{248}/\text{day}$$

Multiple-Server Model (M/M/S Formulas — 1 of 3)

TABLE D.4 Queuing Formulas for Model B: Multiple-Server System (M/M/S)

Parameters & Idle System Probability (P_0)

- M = number of channels (servers) open
- λ = average arrival rate; μ = average service rate at *each* channel

1. Probability that there are zero customers in the system (P_0):

$$P_0 = \frac{1}{\left[\sum_{n=0}^{M-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n\right] + \frac{1}{M!}\left(\frac{\lambda}{\mu}\right)^M \frac{M\mu}{M\mu-\lambda}} \qquad \text{for } M\mu > \lambda$$

Multiple-Server Model (M/M/S Formulas — 2 of 3)

TABLE D.4 Queuing Formulas for Model B: Multiple-Server System (M/M/S)

System Length (L_s) and System Waiting Time (W_s)

2. Average number of customers in the system (L_s):

$$L_s = \frac{\lambda\mu(\lambda/\mu)^M}{(M-1)!(M\mu-\lambda)^2}P_0 + \frac{\lambda}{\mu}$$

3. Average time a customer spends in the system (W_s):

$$W_s = \frac{\mu(\lambda/\mu)^M}{(M-1)!(M\mu-\lambda)^2}P_0 + \frac{1}{\mu} = \frac{L_s}{\lambda}$$

Multiple-Server Model (M/M/S Formulas — 3 of 3)

TABLE D.4 Queuing Formulas for Model B: Multiple-Server System (M/M/S)

Queue Length (L_q) and Queue Waiting Time (W_q)

4. Average number of customers waiting in the queue (L_q):

$$L_q = L_s - \frac{\lambda}{\mu}$$

5. Average time a customer spends waiting in the queue (W_q):

$$W_q = W_s - \frac{1}{\mu} = \frac{L_q}{\lambda}$$

Multiple-Server Example ($\lambda = 2,\ \mu = 3,\ M = 2$ Mechanics)

Step 1: Compute Probability of Zero Cars in the Shop (P_0)

Substitute $\lambda = 2$, $\mu = 3$, and $M = 2$ into the P_0 equation:

$$P_0 = \frac{1}{\left[\frac{1}{0!}\left(\frac{2}{3}\right)^0 + \frac{1}{1!}\left(\frac{2}{3}\right)^1\right] + \frac{1}{2!}\left(\frac{2}{3}\right)^2 \frac{2(3)}{2(3)-2}}$$

$$= \frac{1}{1 + \frac{2}{3} + \frac{1}{2}\left(\frac{4}{9}\right)\left(\frac{6}{4}\right)} = \frac{1}{1 + \frac{2}{3} + \frac{1}{3}} = \frac{1}{2} = \mathbf{0.50}$$

Multiple-Server Example ($\lambda = 2,\ \mu = 3,\ M = 2$ Mechanics)

System Metrics (L_s and W_s)

$$L_s = \frac{2(3)(2/3)^2}{1!(6-2)^2}(0.5) + \frac{2}{3} = \frac{1}{12} + \frac{2}{3} = \mathbf{0.75} \text{ cars}$$

$$W_s = \frac{L_s}{\lambda} = \frac{0.75}{2} = \frac{3}{8} \text{ hr} = \mathbf{22.5} \text{ minutes}$$

Queue Metrics (L_q and W_q)

$$L_q = L_s - \frac{\lambda}{\mu} = 0.75 - 0.667 = \frac{1}{12} = \mathbf{0.083} \text{ cars}$$

$$W_q = \frac{L_q}{\lambda} = \frac{0.083}{2} = 0.0417 \text{ hr} = \mathbf{2.5} \text{ minutes}$$

Economic Impact of Adding a Second Mechanic

Wait Cost: \$160 $\rightarrow$ \$**10**/day | **Labor Cost:** \$88 $\rightarrow$ \$**176**/day | **Total Cost:** \$248 $\rightarrow$ \$**186**/day **(Save \$62/day!)**

Multiple-Server Example: Effect of Service Level

OPERATING CHARACTERISTIC	SINGLE SERVER ($M = 1$)	TWO SERVERS ($M = 2$)
Probability system is empty (P_0)	0.33	**0.50**
Average cars in system (L_s)	2.00 cars	**0.75** car
Average time in system (W_s)	60.0 minutes	**22.5** minutes
Average cars in queue (L_q)	1.33 cars	**0.083** car
Average time in queue (W_q)	**40.0** minutes	**2.5** minutes
Total Daily Cost (Wait + Wage)	\$160 + \$88 = \$248/day	\$**10** + \$**176** = \$**186**/day

Notice the dramatic nonlinear drop in waiting time (40 min → 2.5 min) from adding a second server!

Waiting Line Tables

TABLE D.5 Values of L_q for $M = 1$–5 Servers and Selected Values of λ/μ

λ/μ	$M = 1$	$M = 2$	$M = 3$	$M = 4$	$M = 5$
0.10	0.0111	0.0002	0.0000	0.0000	0.0000
0.25	0.0833	0.0039	0.0002	0.0000	0.0000
0.50	0.5000	0.0333	0.0030	0.0003	0.0000
0.75	2.2500	0.1227	0.0147	0.0018	0.0002
0.90	**8.1000**	**0.2285**	**0.0300**	**0.0041**	**0.0005**
1.00	∞	0.3333	0.0454	0.0067	0.0010
1.60	—	2.8444	0.3128	0.0604	0.0121
2.00	—	∞	0.8888	0.1739	0.0398
2.60	—	—	4.9322	0.6581	0.1609
3.00	—	—	∞	1.5282	0.3541
4.00	—	—	—	∞	2.2164

Waiting-Line Table Example (Bank Teller Staffing)

Problem: $\lambda = 18$ customers/hr, $\mu = 20$ customers/hr per teller $\implies \lambda/\mu = \mathbf{0.90}$

Using Table D.5 at $\lambda/\mu = 0.90$ and Little's Law ($W_q = L_q/\lambda = L_q/18$):

TELLERS (M)	UTILIZATION ($\rho = \frac{\lambda}{M\mu}$)	QUEUE LENGTH (L_q)	WAIT TIME ($W_q = L_q/18$)
$M = 1$	90.0%	8.1000 customers	0.4500 hr $=$ **27.0** min
M = 2	**45.0**%	**0.2285** customer	0.0127 hr $=$ **0.76** min (**45.6** sec)
$M = 3$	30.0%	0.0300 customer	0.0017 hr $=$ **0.10** min (**6.0** sec)
$M = 4$	22.5%	0.0041 customer	0.0002 hr $=$ **0.014** min ($<$ **1** sec)

Constant-Service-Time Model (Model C — M/D/1)

TABLE D.6 Queuing Formulas for Model C: Constant-Service-Time System (M/D/1)

Queue Metrics (Cut in Half vs. M/M/1!)

1. Average length of the queue (L_q):

$$L_q = \frac{\lambda^2}{2\mu(\mu - \lambda)}$$

2. Average waiting time in the queue (W_q):

$$W_q = \frac{\lambda}{2\mu(\mu - \lambda)}$$

System Metrics (L_s and W_s)

3. Average customers in the system (L_s):

$$L_s = L_q + \frac{\lambda}{\mu}$$

4. Average time in the system (W_s):

$$W_s = W_q + \frac{1}{\mu}$$

*Eliminating service-time variability cuts both L_q and W_q by exactly **50%** compared to exponential service!*

Constant-Service-Time Example (Garcia-Golding Recycling)

Problem Data

- Truck arrival rate: $\lambda = \mathbf{8}$ trucks/hr
- Automated compactor rate: $\mu = \mathbf{12}$ trucks/hr (constant 5 min unload)
- Truck + driver waiting cost: \$**60**/hr
- Current manual wait time: **15** min ($\frac{1}{4}$ hr)
- New compactor cost: \$**3** per truck unloaded

M/D/1 Wait Time & Savings Analysis

- **New queue wait (W_q):**

$$W_q = \frac{\lambda}{2\mu(\mu - \lambda)} = \frac{8}{2(12)(4)} = \mathbf{\frac{1}{12}} \text{ hr } (\mathbf{5} \text{ min})$$

- **Current wait cost/trip:** $\$60 \times \frac{1}{4}$ hr $= \$15$/trip
- **New wait cost/trip:** $\$60 \times \frac{1}{12}$ hr $= \$5$/trip
- **Wait savings/trip:** $\$15 - \$5 = \$10$/trip
- **Net savings/trip:** $\$10 - \$3$ (equipment) $=$ \$**7**/trip

Little's Law

Fundamental Conservation Law of Flow Systems

$$\mathbf{L_s} = \lambda \mathbf{W_s} \qquad \text{and} \qquad \mathbf{L_q} = \lambda \mathbf{W_q}$$

Scope & Universality

- Holds for **any steady-state queuing process** regardless of arrival/service distributions, servers, or priority rules.
- Applies to all queuing models **except the finite-population model** (where the effective arrival rate is $(N - L_s)\lambda$).

Managerial Value

- Knowing any **two** of (L, λ, W) immediately determines the third.
- Connects **Work-in-Process inventory (L)** directly to **Flow / Lead Time (W)**.

Little's Law Example

Coffee Shop Queue Analysis

- Suppose a coffee shop has an average arrival rate of $\lambda = \mathbf{20}$ customers per hour.
- By simple visual observation, there are on average $\mathbf{L_q} = \mathbf{5}$ customers waiting in line.
- **Question:** How long does the average customer wait in line (W_q)?

$$W_q = \frac{L_q}{\lambda} = \frac{5 \text{ customers}}{20 \text{ customers/hour}} = \mathbf{0.25} \text{ hour} = \mathbf{15} \text{ minutes}$$

Applications of Little's Law

1. Department Store Occupancy

- Shoppers enter at $\lambda = 220$ shoppers/hr.
- Average time spent in store: $W = 45$ min $= 0.75$ hr.
- **Average shoppers in store (L):**

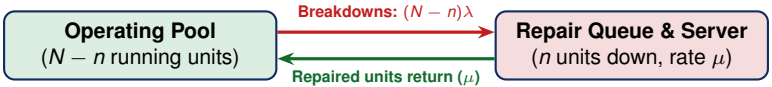
$$L = \lambda W = 220 \times 0.75 = \textbf{165} \text{ shoppers}$$

2. Drug Store Inventory Turnover

- Store holds average inventory $L = 80$ boxes.
- Sales rate: $\lambda = 240$ boxes/month $= 8$ boxes/day (30-day month).
- **Average time a box sits in inventory (W):**

$$W = \frac{L}{\lambda} = \frac{80 \text{ boxes}}{8 \text{ boxes/day}} = \textbf{10} \text{ days}$$

Finite-Population Model (Model D)



Closed-Loop Finite-Population Characteristics

- **Limited population (N):** Potential customers form a small, closed pool (e.g., $N = 5$ printers or factory machines).
- **State-dependent arrival rate:** When n units are down, only $N - n$ operating units can break down, so the effective arrival rate is $(N - n)\lambda$ (average $(N - L_s)\lambda$).
- **Assumptions:** Poisson breakdown rate λ per operating unit, exponential repair rate μ, single server ($M = 1$), FIFO.

Finite-Population Model (Table D.7 — 1 of 2)

TABLE D.7 Queuing Formulas for Model D: Finite-Population System

Parameters: λ = arrival rate per unit, μ = service rate, N = population size

1. Probability that the system is empty (P_0):

$$P_0 = \frac{1}{\sum_{n=0}^{N} \frac{N!}{(N-n)!} \left(\frac{\lambda}{\mu}\right)^n}$$

2. Average waiting time in the queue (W_q) and in the system (W_s):

$$W_q = \frac{L_q}{(N - L_s)\lambda} \qquad W_s = W_q + \frac{1}{\mu}$$

Finite-Population Model (Table D.7 — 2 of 2)

TABLE D.7 Queuing Formulas for Model D: Finite-Population System

Queue Length (L_q), System Length (L_s), and State Probability (P_n)

3. Average length of the queue (L_q):

$$L_q = N - \left(\frac{\lambda + \mu}{\lambda}\right)(1 - P_0)$$

4. Average number of units in the system (L_s):

$$L_s = L_q + (1 - P_0)$$

5. Probability of n units in the system (P_n):

$$P_n = \frac{N!}{(N-n)!}\left(\frac{\lambda}{\mu}\right)^n P_0 \qquad \text{for } n = 0, 1, 2, \ldots, N$$

Finite-Population Example (1 of 3)

Problem: Department of Commerce ($N = 5$ Laser Printers)

- **Population:** $N = 5$ printers; each breaks down every 20 operating hours ($\lambda = \mathbf{0.05}$ printer/hr).
- **Repair capacity:** Single technician repairs a printer in 2 hours ($\mu = \mathbf{0.50}$ printer/hr $\implies \lambda/\mu = 0.10$).

Step 1 & Step 2: Compute P_0 and L_q

$$P_0 = \frac{1}{\sum_{n=0}^{5} \frac{5!}{(5-n)!}(0.10)^n} = \frac{1}{1 + 5(0.1) + 20(0.01) + 60(0.001) + 120(0.0001) + 120(0.00001)} = \frac{1}{1.7732} = \mathbf{0.564}$$

$$L_q = N - \left(\frac{\lambda + \mu}{\lambda}\right)(1 - P_0) = 5 - \left(\frac{0.05 + 0.50}{0.05}\right)(1 - 0.564) = 5 - 11(0.436) = \mathbf{0.20} \text{ printer}$$

Finite-Population Example (2 of 3)

Steps 3, 4, and 5: Printers Down (L_s) and Turnaround Times (W_q, W_s)

- **3. Average number of printers in the repair system (L_s):**

$$L_s = L_q + (1 - P_0) = 0.20 + (1 - 0.564) = \mathbf{0.64} \text{ printer}$$

- **4. Average waiting time in the repair queue (W_q):**

$$W_q = \frac{L_q}{(N - L_s)\lambda} = \frac{0.20}{(5 - 0.64)(0.05)} = \frac{0.20}{0.218} = \mathbf{0.91} \text{ hour } (\mathbf{55} \text{ minutes})$$

- **5. Average total time a broken printer spends in the system (W_s):**

$$W_s = W_q + \frac{1}{\mu} = 0.91 + \frac{1}{0.50} = 0.91 + 2.00 = \mathbf{2.91} \text{ hours}$$

Finite-Population Example (3 of 3): Total Hourly Cost

Cost Parameters

- Printer downtime cost: \$**120** per hour per broken printer
- Technician wage rate: \$**25** per hour
- Average number of printers down (L_s): **0.64** printer

Total Hourly System Cost

$$\begin{aligned}\text{Total hourly cost} &= (L_s \times \text{Downtime cost/hr}) + \text{Technician wage/hr} \\ &= (0.64)(\$120/\text{hr}) + \$25/\text{hr} \\ &= \$76.80 + \$25.00 = \$\mathbf{101.80} \text{ per hour}\end{aligned}$$